

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0532	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)		
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Section / Batch:	AIML/2024	Date:	06/08/2026

Code of Conduct

I K.Vijay Bhaskar Reddy(192425155)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

K.Vijay Bhaskar Reddy

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation is an image enhancement technique that modifies the intensity (gray) values of pixels in an image. It improves the visual appearance of an image by changing its brightness and contrast.

Applications:

Enhances image contrast.

Improves image brightness.

Highlights important image details.

Makes hidden features visible.

Prepares images for further image processing tasks such as segmentation and feature extraction.

(b) Negative Transformation

Formula:

$$s = (L - 1) - r$$

Where:

r = Original pixel value

s = Transformed pixel value

L = Total number of gray levels

For an 8-bit image, $L = 256$, therefore:

$$s = 255 - r$$

Example:

Original Pixel (r)	Negative Pixel (s)
20	235
50	205
100	155
150	105
200	55
255	0

Transformed Values = [235, 205, 155, 105, 55, 0]

Program 1: Grey Level Negative Transformation

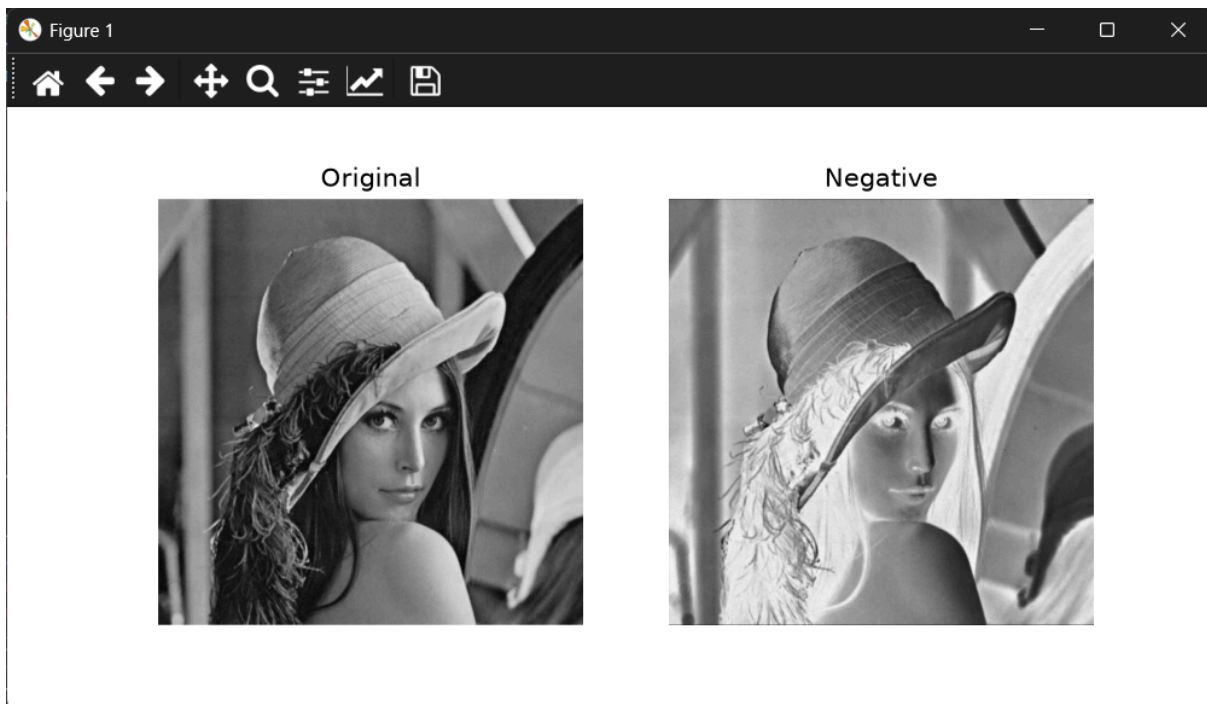
```
import cv2
import matplotlib.pyplot as plt

# Read image in grayscale
img = cv2.imread("image.jpg", 0)

# Negative transformation
negative = 255 - img

# Display
plt.subplot(1,2,1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(negative, cmap="gray")
plt.title("Negative")
plt.axis("off")
plt.show()
```



2. Log Transformation

(a) How Log Transformation Enhances Low-Intensity Pixels

Log transformation is represented by:

$$s = c \log(1 + r)$$

This transformation expands low-intensity (dark) pixel values while compressing high-intensity (bright) values. As a result, details present in dark regions become more visible without excessively increasing the brightness of already bright regions.

Advantages:

- Enhances dark regions of an image.

- Compresses high intensity values.

- Improves visibility of hidden details.

- Commonly used in medical and satellite image processing.

(b) Computation

Given: $c = 1$ and $\log$ is the natural logarithm ($\ln$).

Input Pixel (r)	Output $s = \ln(1+r)$
0	0.000
1	0.693
3	1.386
7	2.079
15	2.773

Program 2: Log Transformation

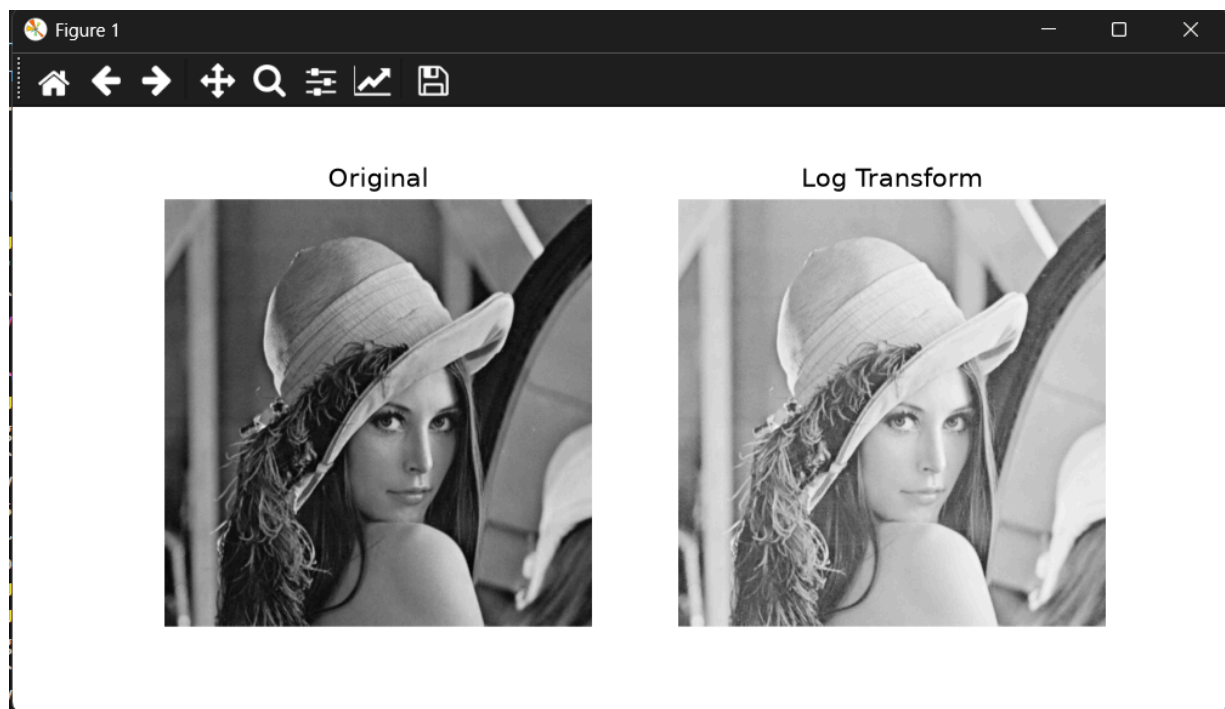
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)

c = 255 / np.log(1 + np.max(img))
log_img = c * np.log(1 + img.astype(np.float32))
log_img = np.array(log_img, dtype=np.uint8)

plt.subplot(1,2,1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(log_img, cmap="gray")
plt.title("Log Transformation")
plt.axis("off")
plt.show()
```



3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization is a contrast enhancement technique that redistributes the gray levels of an image so that the histogram becomes more uniformly distributed.

Benefits:

- Improves image contrast.
- Enhances visibility of details.
- Uses the complete range of gray levels.
- Produces a more balanced brightness distribution.

(b) Histogram Equalization

Given Histogram (3-bit Image):

Intensity	Frequency
0	2
1	2
2	4
3	8

Total Pixels (N) = 16

Gray Levels (L) = $2^3 = 8$

Step 1: Cumulative Distribution Function (CDF)

Intensity	Frequency	Cumulative Frequency	CDF	Equalized Intensity
0	2	2	0.125	1

1	2	4	0.250	2
2	4	8	0.500	4
3	8	16	1.000	7

Formula Used:

$$\text{Equalized Intensity} = (L - 1) \times \text{CDF} = 7 \times \text{CDF}$$

Final Equalized Intensity Mapping:

Original Intensity	Equalized Intensity
0	1
1	2
2	4
3	7

Program 3: Histogram Equalization

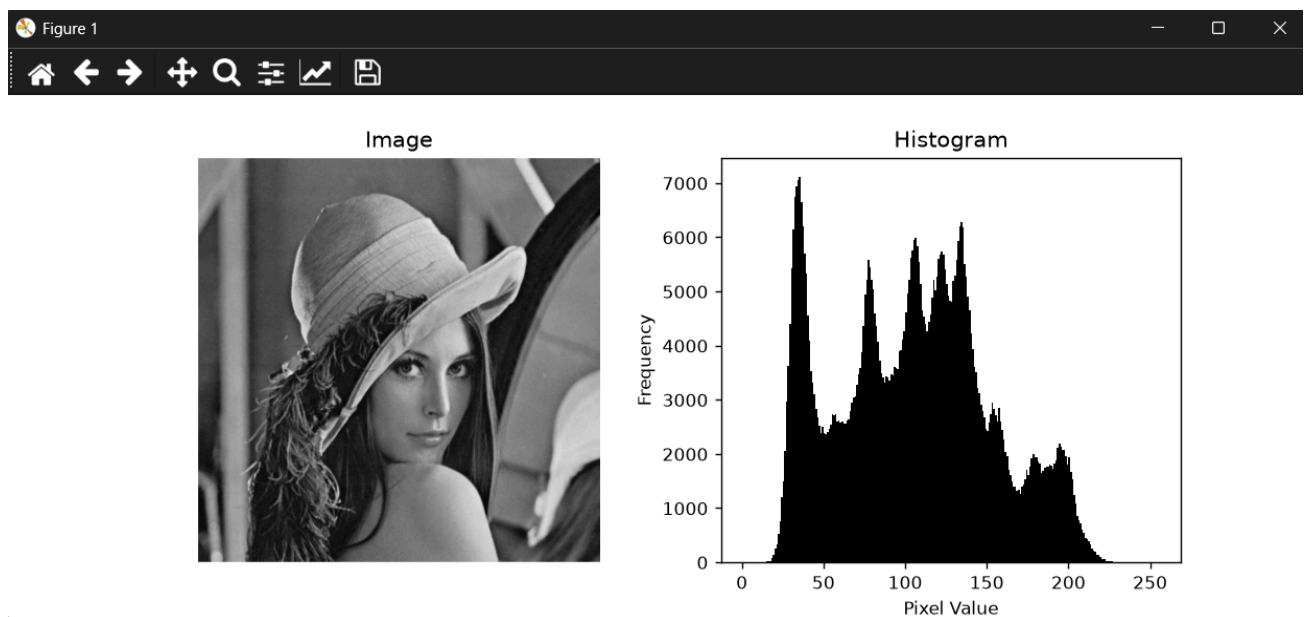
```
import cv2
import matplotlib.pyplot as plt

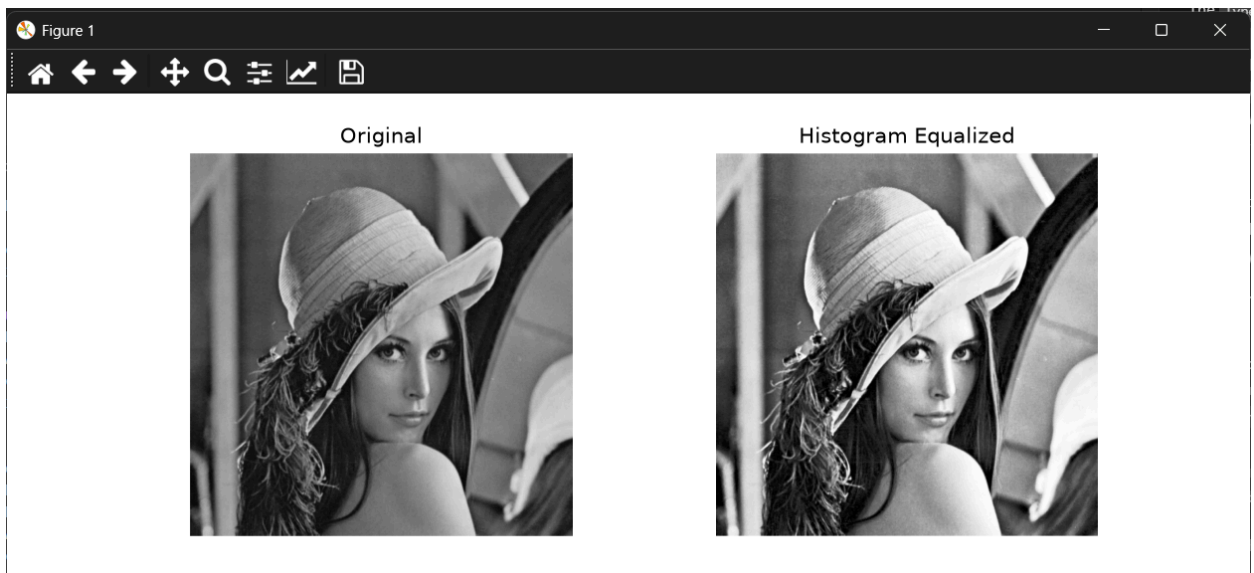
img = cv2.imread("image.jpg", 0)

equalized = cv2.equalizeHist(img)

plt.subplot(1,2,1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")
plt.show()
```





Conclusion

Grey level transformation, log transformation, and histogram equalization are fundamental image enhancement techniques. Grey level transformation modifies brightness, log transformation improves details in dark regions, and histogram equalization enhances overall image contrast by redistributing intensity values.